CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

2^{nd} Semester of B. Pharm. (OLD) Examination
University Theory Examination April/May 2015
MA-141, Advanced Mathematics
Date: 20/05/2015 (Wednesday) Time: 10.00 a.m to 1.00 p.m Maximum Marks: 80

INSTRUCTIONS:

(a) Figures to right indicate marks.
(b) Section wise separate answer book to be used.
(c) Draw neat sketches wherever necessary.

SECTION-I

Q.1 Attempt any two: [20]

(1)(a) Find n^{th} derivative of e^{2x}. [03]

(b) Find Laplace transform of $e^{2t} - \sin t + \cos 5t$. [03]

(c) Find Maclaurin's series expansion of e^{x}. [04]

(2)(a)(i) State Cauchy's mean value theorem. [02]

(ii) True or False: Every Macluarin's series is Taylor series. [01]

(b) Find the inverse Laplace transform of $\frac{1}{s} - \frac{1}{s^2+1} + \frac{1}{s-7}$. [03]

(c) Find 11^{th} derivative of $e^{3x}\cos(4x-5)$. [04]

(3)(a)(i) Define: Taylor's series expansion of a function. [02]

(ii) True or False: Lagrange's mean value theorem can be apply for discontinuous function also. [01]

(b) Find 3^{rd} derivative of $\sin(7x+3)$. [03]

(c) Find Laplace transform of $e^{t}(\sin 2t - \cos 4t)$. [04]

Q.2 Attempt any two: [20]

(1)(a) Find n^{th} derivative of $\frac{1}{(x-1)(x+1)}$. [05]

(b) Find n^{th} derivative of x^2e^x using Leibnitz's theorem. [05]

(2)(a) Find the inverse Laplace transform of $\frac{1}{s(s-9)}$. [05]

(b) Solve the differential equation $\frac{dy}{dt} - y = 1$ with initial condition $y(0) = 0$ using Laplace transform. [05]

(3)(a) Verify Lagrange's mean value theorem for $f(x) = x^2$ in $[0, 2]$. [05]

(b) Expand $x^2 + x + 9$ in powers of $(x + 3)$. [05]

SECTION-II

Q.3 Attempt any two: [20]

(1)(a) Solve the differential equation $\frac{dy}{dx} + y = e^x$. [05]

(b) Solve the differential equation $\sin y\ dx + x \cos y\ dy = 0$. [05]

(2)(a) Solve the differential equation $(D^2 - 4)y = e^{-2x}$ using method of variation of parameters. [06]

(b) What is order and degree of differential equation $(\frac{d^3y}{dx^3})^2 - \frac{dy}{dx} + y = e^x$? [02]

(c) What is auxiliary equation of differential equation $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 9y = 0$? [02]

(3)(a) Solve the differential equation $\frac{d^2y}{dx^2} - y = 1$. [05]

(b) Solve the linear differential equation $\frac{dy}{dx} = x + 2$. [05]

Q.4 Attempt any two: [20]

(1)(a) Solve the differential equation $\frac{dy}{dx} = \frac{y}{e^x}$ [02]

(b) Solve the differential equation $(D^2 - 9)y = 0$ [02]

(c) The rate at which bacteria multiply is proportional to the instantaneous number present. If the original number doubles in 2 hours, in how many hours will it triple? [06]

(2)(a) If the temperature of a body drops from 100° to 60° in one minute when the temperature of the surrounding is 20°, what will be the temperature of the body at the end of second minute? [06]

(b) Solve the differential equation $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 4y = 0$ [04]

(3)(a) Bacteria multiply at the rate proportional to the number present. If the original number N doubles in three hours, find in how many hours, the numbers of the bacteria will be $4N$? [06]

(b) Solve the differential equation $xy^2\, dx + x^2y\, dy = 0$ [04]

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